ADVANCED MACHINE LEARNING · PROJECT 21 · PHASE 1

Will your flight be late tomorrow?

Predicting flight delays the night before departure

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github.com/techatanu/Flight_Delay_Risk

The question, the rule and the data

The question

Tonight, can we tell which of tomorrow's flights will land 15 or more minutes late?

The rule

Use only what is known tonight: tomorrow's schedule and how earlier days went. Nothing from the flight day itself.

336,776

flights from New York's three airports in 2013

1 in 4

arrived 15+ minutes late

9,430

cancelled or diverted flights left out

Removed

departure delay: it is only known after take-off, so using it would be cheating

Three things the data told us

10% → 37%

Delays build up during the day

Early-morning flights are rarely late. Evening flights inherit the day's delays, because a late plane makes its next flights late too.

34% vs 18%

December is the worst month

October and November, the months we practised on, were the calmest. This comes back later.

Carry-over

Bad days repeat

A day with many delays is a warning for the next day, so recent delays are a useful clue.

35 clues, all known the night before

When

time of day, weekday, nearby holidays

What

route, distance, airline

How busy

flights planned at the airport that day and hour

Track record

past delay rate of the route, flight number, airline and aircraft

Recent days

delays over the last week and yesterday

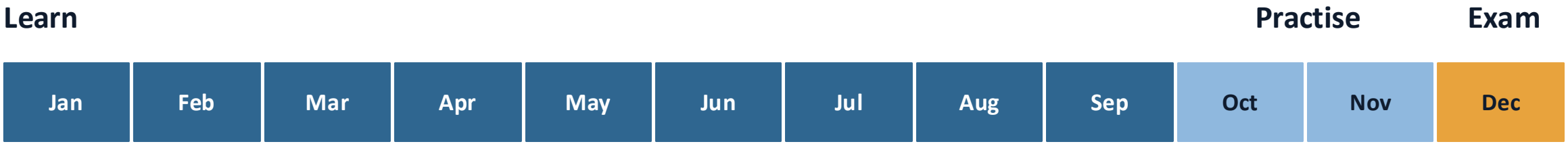
No-peeking test

We scrambled every outcome after a chosen date and rebuilt all the clues.

If a clue for an earlier day had changed, it would have been secretly using the future.

Nothing changed: test passed

A fair test: learn, practise, then one final exam



All choices were made on the practice months. December was used only once.

Logistic regression

A simple model: it weighs each clue and adds up the evidence.

XGBoost

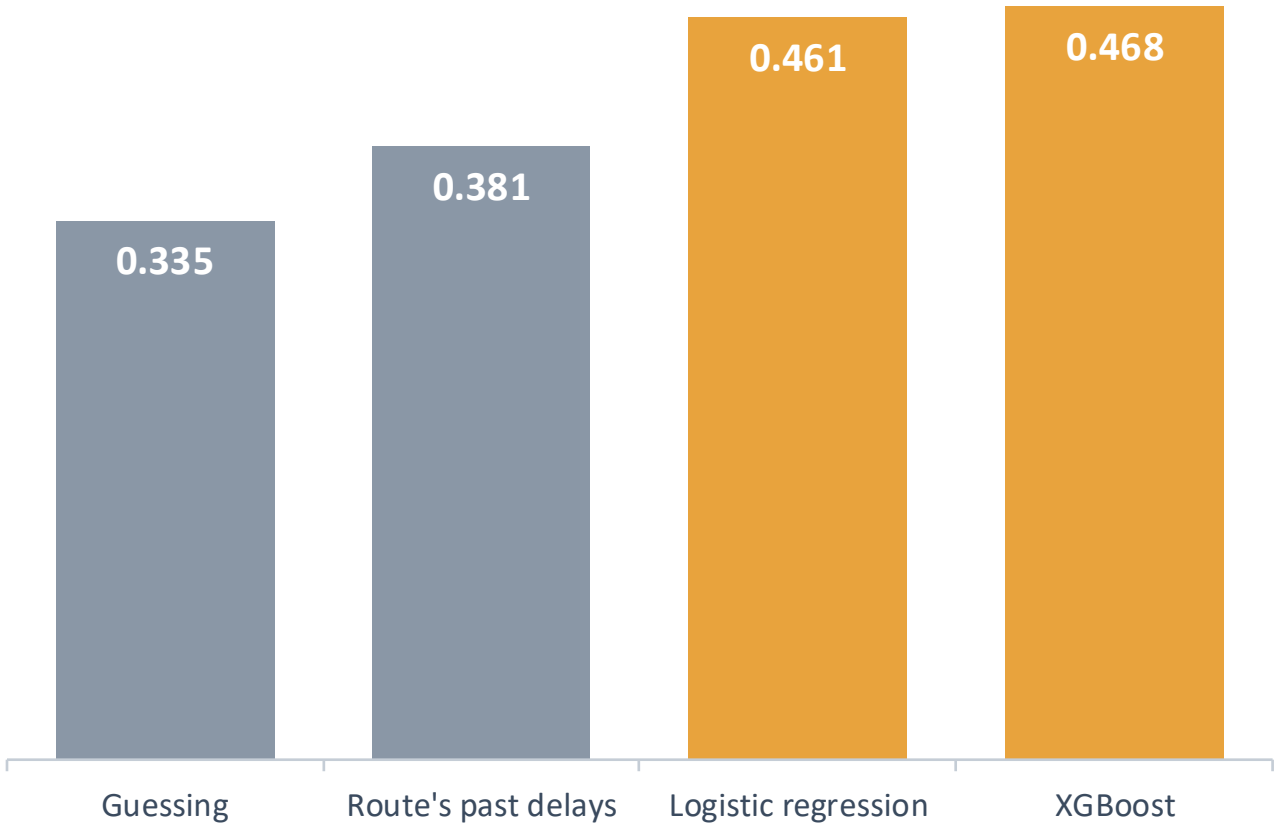
Hundreds of small decision trees, each correcting the ones before it.

What XGBoost taught us

At first it memorised individual days instead of learning general patterns. We fixed this by making it learn only from large groups of at least 8,000 flights.

Both models beat simple guessing

How well each method ranks risky flights in December (higher is better)



66 of 100

flights that XGBoost flags as high-risk really were late

The catch

Without tomorrow's weather, which we don't have, only a small share of late flights can be flagged that confidently.

December surprised the model, so we let it adjust

23%

late flights the model expected in December

34%

late flights that really happened

Why?

The model learned what "normal" looks like from the calm practice months, October and November.

Our fix: every day, the model re-adjusts its probabilities using the real outcomes of the last 4 weeks, which are known the night before.

0.102 → 0.047

probability error cut by more than half

Almost 2×

as many late flights flagged (3.5% → 6.2% of them)

Could moving a flight help? Only a little

27,020
December flights

1,979
had high delay risk

133
improved when moved

98
of those moved earlier

The rule we tested

Try moving each high-risk flight one or two hours earlier or later on the same day. Keep the move only if its risk clearly drops.

The lesson

Delay risk comes mostly from the day and the route, not the exact hour. Moving earlier helps a little.

A what-if study: it ignores crews and passenger connections.

What we learned

- Delays can be predicted a day ahead without peeking at the future
- Our models clearly beat guessing, even without weather data
- Re-adjusting every day keeps the model reliable when the season changes
- Next, in Phase 2: a deep learning model on the same data and the same test

Atanu Adhikari
Data exploration, slides

Sambhav Kumar
Clues (features), README

Nishant Ranjan Singh
Model training, report

Sankalp M Tellur
Results and evaluation

Prince Singh
Rescheduling, report

Thank you